Studying and Enhancing Operational Efficiency at Mahak Spices Company

A Final report for the BDM capstone Project

Submitted by

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1 Executive Summary

Mahak Spices functions as a B2B spice trading business in Subhash Nagar, Shahjahanpur, Uttar Pradesh. Providing a broad variety of spices that has helped local retailers and also small businesses. The business has built a strong local presence within. However, the business is indeed overdependent upon a few high-demand spices, its perishable inventory spoils in the summer months as its transportation costs are highly inconsistent because of these three key operational challenges. These factors have limited profitability with overall efficiency.

To tackle the identified challenges, a comprehensive data-driven review was carried out on the shop's daily records of sales, purchases, and transport from the past six months. The review was focused on isolating slow-moving spices, measuring the revenue of each product, mapping monthly profit patterns, and evaluating fluctuations in shipping costs. The data was systematically cleaned, organized, and analyzed using Excel and Python to derive actionable insights.

Based on the analysis, a set of recommendations has been developed to improve efficiency and profitability. These include more effective utilization of inventory, reduction of losses from spoilage, and resolution of inconsistencies in logistics costs. By identifying clear performance trends in various spice categories and recognizing seasonal patterns in sales and spoilage, this report addresses immediate inefficiencies and prepares them for long-term operational improvements. The insights gained can help the business take a more forward-thinking and responsive approach to managing inventory and logistics. This way, Mahak Spices can stay competitive and sustainable in the changing B2B market.

2 Detailed Explanation of Analysis/Methods

To address the three main operational challenges faced by Mahak Spices, including reliance on a small range of spices, spoilage of inventory during the summer, and fluctuating transportation costs, a detailed data analysis process was carried out. The steps below describe the methods and techniques used:

2.1 Data Collection

The first step was to gather daily data from Mahak Spices over six months, from November 2024 to April 2025. The dataset included:

- Total daily sales and purchase values
- The quantity sold of each spice per day
- The cost of each spice per day
- Daily transportation costs

This data formed the basis for all further analysis, which aimed to find patterns and inefficiencies in the business operations.

2.2 Data Cleaning and Formatting

Before analysis, the data was cleaned and standardized to ensure accuracy:

- Missing values (e.g., no spice sold on a particular day) were replaced with zeroes where appropriate.
- Incorrect or inconsistent entries were checked, especially for numeric and date fields.
- Column headers were renamed using consistent formats (e.g., Turmeric_sold, RedChilli_cost_per_kg) for clarity.
- Data types were standardized (like date, currency, number etc).
- Duplicate rows and anomalies (like negative sales or transport cost) were reviewed and corrected.

This ensured the dataset was free from inconsistencies and ready for meaningful analysis.

2.3 Data Structuring and Feature Engineering

It is a critical step in the data analysis process where raw data is transformed into a more organized, meaningful, and analysis-ready format. This step involves both reorganizing existing data and creating new variables (or "features") that provide deeper insights into business operations. To derive actionable insights, several new variables and columns were calculated:

- Revenue per product per day = Quantity sold × Selling price
- Profit per product per day = Revenue – Cost
- Total daily profit = Sum of profits across all spices

This step helped quantify the financial contribution of each spice and provided a basis for month-wise performance assessment.

2.4 Descriptive Statistical Analysis

To understand the basic characteristics of the data and detect irregularities:

- Mean, median, and standard deviation were calculated for each spice's daily sales.
- These statistics helped identify high variability in sales (e.g., Red Chilli showing large standard deviation due to festival spikes).
- Skewness was also evaluated to understand whether sales were centred around average values or dominated by outliers.
- This analysis helped address the first challenge: overdependence on certain spices, as it clearly revealed which products were consistently contributing to revenue and which were underperforming.

2.5 Correlation Analysis

Correlation analysis is a statistical method used to measure the strength and direction of the relationship between two variables. The most commonly used metric is the Pearson correlation coefficient, which ranges from -1 to +1. Correlation analysis was performed to examine relationships between sales of different spices and between sales and transport costs.

Steps to perform correlation analysis-

- Calculate Pearson's coefficient

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$$

- Results - if the value comes out to be positive it indicates positive correlation i.e, the values of both are variables are showing the same trend while negative correlation indicates different trends.
- To visualize plot heatmaps for the sales of different spices.
- From this analysis the sales dependencies across various spices and transportation costs can be determined. By calculating Pearson's correlation coefficient, we can

identify whether the sales of one spice are influenced by another, and whether daily transportation expenses are linked to the volume of goods sold.

2.6 ABC Classification

ABC Analysis is an inventory categorization technique that divides products into three categories A, B, and C based on their contribution to total revenue or value. It is based on the Pareto Principle (80/20 Rule), which states that a small percentage of items often contribute to a large portion of the business value.

Steps to perform ABC analysis –

- Rank spices by calculating total revenue for each spice
- Sort these spices in descending order and calculate their percentage contribution
- Categorize the spices into three categories
    - A – High Value items (70 % of the sales)
    - B – Moderate Value items (15 % of the sales)
    - C – Low Value items (5 % of the sales)
- To visualize plot pareto or bar charts to represent each category
- It highlights which spices contribute most to revenue, allowing the business to prioritize stocking high-performing items. This helps reduce spoilage from slow-moving products and improves overall profitability.

2.7 Trend and Seasonal Analysis

Trend analysis focuses on the long-term direction of data whether it increases, decreases, or remains stable. Seasonal analysis identifies short-term, recurring patterns that repeat at regular intervals like months or quarters. Together, they help in understanding, forecasting, and planning business activities.

Steps to perform trend and seasonal analysis –

- Group monthly sales data and plot line charts for each spice.
- Identify the months with high and low sales of each product

- In order to track spoilage indicators, compare the summer months (March-April) with other months.
- With this analysis the seasonal demands and product spoilage trends can be identified.

2.8 Profitability and Transport Cost Analysis

Profitability analysis is the process of assessing how much profit each product or activity contributes to the business. It helps identify which items are driving revenue and which are causing losses. Transportation cost analysis involves evaluating logistics expenses to understand cost patterns and efficiency. It helps identify inconsistencies in daily delivery charges.

After plotting daily and monthly profit trends, we can compare profitability with transport trends and get insights into how fluctuating transport costs affect margins, guiding better logistics decisions.

2.9 Software Tools Used

Two main data analysis tools utilized in this project were Microsoft Excel and Python.

- Microsoft Excel was utilized for data formatting, structuring, and cleaning. It facilitated the organization of raw data, dealing with missing values, standardizing column headers, and performing preliminary calculations like profit and revenue. Excel was also helpful in making rapid summaries and tabular analysis.
- **Python**, along with its powerful libraries **Pandas**, **Matplotlib**, and **Seaborn**, was employed for sophisticated data processing and visualization. **Pandas** enabled efficient handling and manipulation of large datasets, while **Matplotlib** and **Seaborn** were used to generate a variety of insightful visualizations including line charts, bar charts, pie charts, and heatmaps. These visuals effectively showcased sales trends, profit margins, and seasonal patterns. Collectively, these tools streamlined the entire workflow from raw data to actionable business insights.

3. Result and Findings

The main conclusions drawn from the statistical and exploratory study of Mahak Spices business operations, which was carried out with Python and Excel, are presented in this part.For reproducibility and transparency, each visualization was created in an interactive Google Colab notebook.

Colab Notebook: **View Full Analysis Here**

3.1 Product Performance and Revenue Contribution

- Turmeric stands out as the top contributor, making up 42.4% of the total revenue. This clearly shows how much customers rely on it, and it highlights Turmeric's importance to the business's overall success.
- Red Chilli comes in second with a strong 33.3% share, showing that despite being perishable especially in the summer it continues to be a high-demand, high-value product.
- Together, Turmeric and Red Chilli account for nearly 76% of all revenue. While that's great for short-term profit, it also means the business is heavily dependent on just two products. Any issue with their supply or quality could create a serious dent in earnings.
- Dhaniya (coriander) contributes a respectable 15.5%, making it a stable, mid-level performer. With the right push like seasonal promotions or better visibility it could help balance out the overreliance on Turmeric and Red Chilli.
- On the other hand, Fennel (Saunf), Ajwain, and Cumin (Jeera) contribute 6.0%, 2.1%, and just 0.8% respectively. Their impact is minimal, and they may be taking up more space and investment than they return.
- This suggests it's time to rethink how these low-contributing spices are managed. Whether it's by reducing stock, bundling them with more popular products, or offering limited-time deals, these items need a fresh strategy.
- Overall, the business could benefit greatly by doubling down on what's working Turmeric, Red Chilli, and Dhaniya while reworking its approach to slower-moving spices.

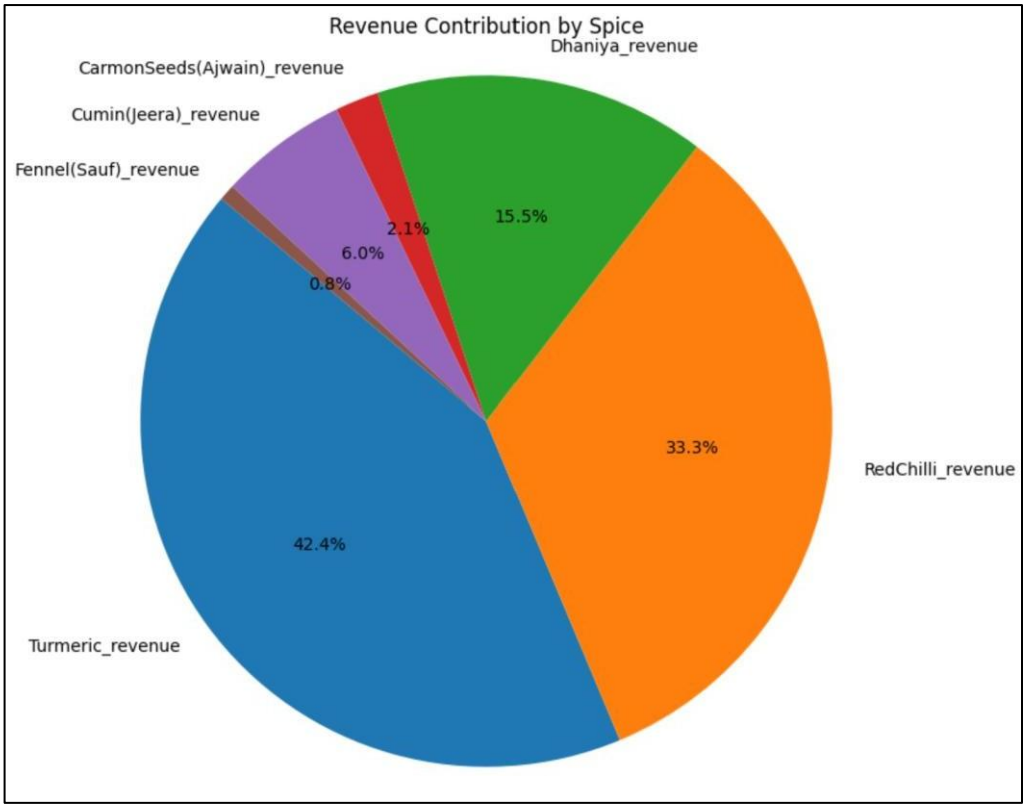

Fig 1 – Pie Chart: Revenue Contribution by Spice

Based on this these spices can be segmented into Category A, Category B and Category C

- **Category A – High Revenue Contributors**

  These are the most critical spices for the business in terms of revenue. They should be prioritized in inventory planning, restocking and promotion.

  - Turmeric
    - Highest revenue contributor (over ₹9,00,000+).
    - Drives the overall business performance, should be monitored closely for supply pricing and demand.
  - Red Chilli
    - Second highest contributor (approx. ₹7,50,000+).
    - It's a key driver of sales especially important to manage well in summer months to reduce spoilage.

- **Category B – Moderate Revenue Contributors**

  These products contribute a moderate share to revenue. They offer potential growth with proper pricing strategies.

  - Dhaniya
    - Contributes significantly, though well behind Category A items.

    - Approximate revenue is ₹3,50,000.

- **Category C – Low Revenue Contributors**

  These spices have minimal impact on total revenue. They require cost-effective inventory management or potential reevaluation. These include – Cumin, Carmon Seeds and Fennel.

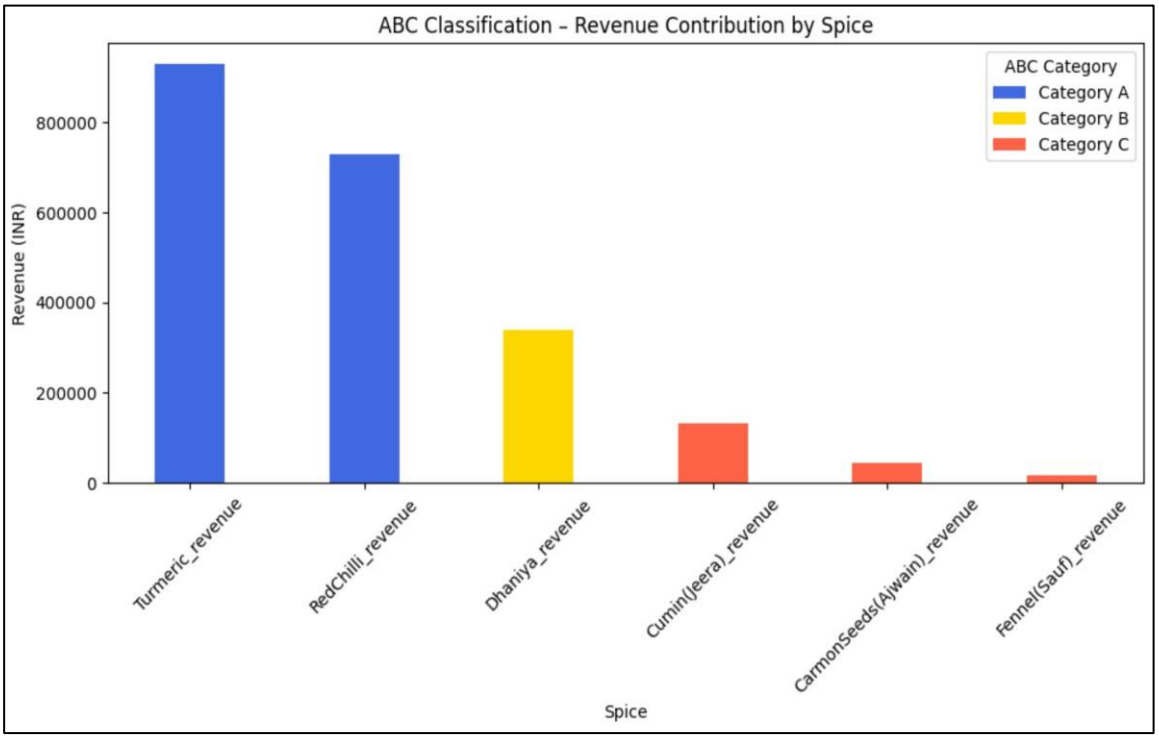

Fig 2 – Bar Chart: ABC Classification of Spices by Revenue

3.2 Monthly Spice Sales and Seasonal Trends

- Turmeric shows steady growth across all observed months, with a clear upward trend in sales volume. This consistency highlights Turmeric as a cornerstone product for the business. Its high demand, especially during peak months like February and March, indicates strong customer dependency and minimal volatility, making it a stable and profitable item to invest in and prioritize during inventory planning.
- Red Chilli displays solid sales in the initial months (November to February), peaking in February with impressive quantities sold. However, this is followed by a steep decline in March and April, which may be attributed to spoilage due to heat and poor storage conditions during the summer months. This pattern

reinforces the need for better perishability management, including cold storage solutions or dynamic pricing to clear stock before spoilage.

- Dhaniya begins with moderate sales, showing a gradual decline until January. However, the trend reverses sharply from February onward, with a marked increase in March and April. This resurgence suggests seasonal or festive demand, potentially linked to specific culinary needs during spring festivals or harvest-related consumption trends. The spice may benefit from targeted marketing during these months to boost returns.
- Cumin (Jeera) exhibits a sudden and notable sales spike in January, which deviates from its otherwise low-volume trend. This anomaly could be due to a bulk order, a localized event, or a specific seasonal preparation requirement. Identifying the cause of this spike can help the business replicate such opportunities in the future.
- Ajwain and Fennel (Saunf) consistently report the lowest sales volumes across all months, indicating that they are not core revenue drivers for the business. Their continued presence in inventory may tie up resources unnecessarily, and a reassessment of their stock levels, pricing strategy, or bundling potential should be considered to avoid waste and improve space utilization.

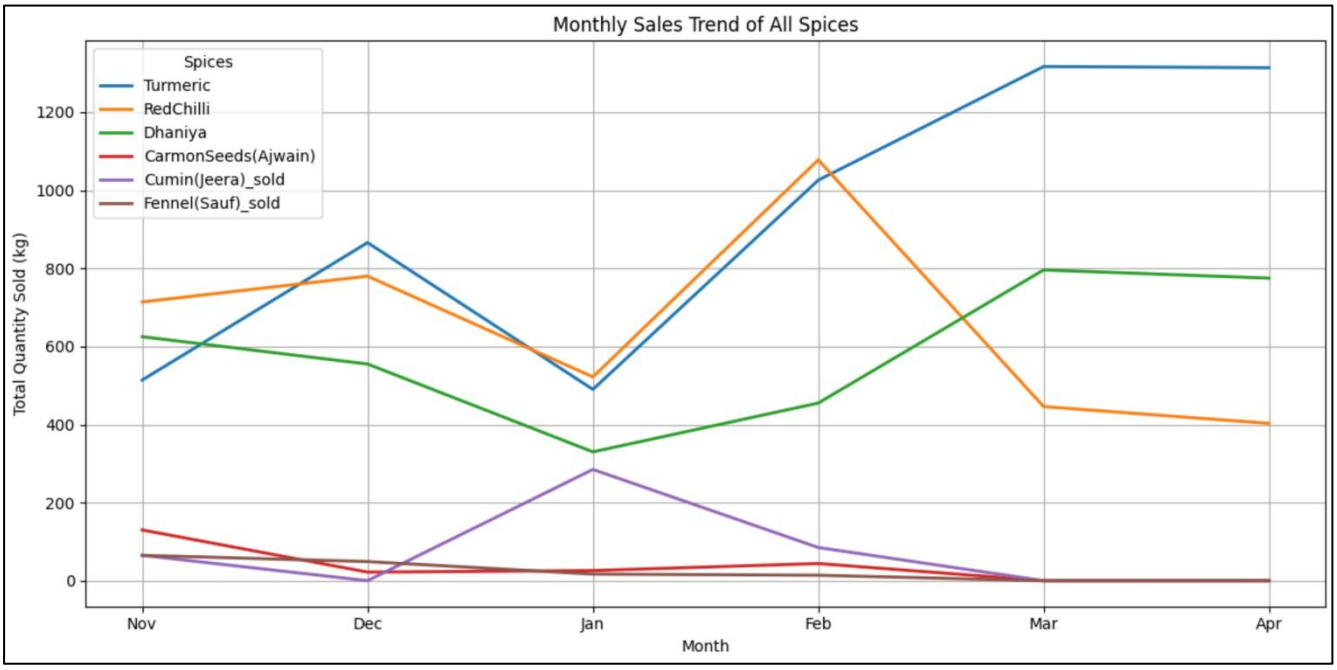

Fig 3 – Line Chart: Monthly Sales Trend of All Spices

3.3 Transport Cost and Logistics Analysis

- This scatter plot shows the relationship between the total quantity of spices sold and the transportation costs for those days.
- The red regression line clearly indicates an upward trend. As sales increase, transportation costs also tend to rise. This is a common pattern in logistics.
- However, the data points vary widely, especially around mid and high sales volumes. This suggests that transport costs do not always directly match the quantity sold. Other factors, like delivery location, fuel prices, or shipment frequency, may be affecting these differences.
- Some data points indicate high transport costs even with moderate sales. This highlights inefficiencies or inconsistencies in shipment practices. These outliers point to areas where costs may be leaking.
- The shaded area around the trend line shows some uncertainty in the prediction, emphasizing the variability in transport expenses, even with similar sales levels.
- The positive slope of the trend line confirms a general correlation, but the strength of this correlation may be moderate at best. This highlights the need for better transport planning.

Fig 4 – Scatter Plot with Trend Line: Total Quantity Sold vs Transport Cost

- The bar chart shows the total transport expenses for each month.
- March has the highest transport cost, which might suggest a surge in operations or inefficient logistics at that time.
- There is a consistent rise from January to March, indicating an increasing transportation burden as summer nears.
- February and April have similar values, but April sees a small decline, possibly due to changes in operations.

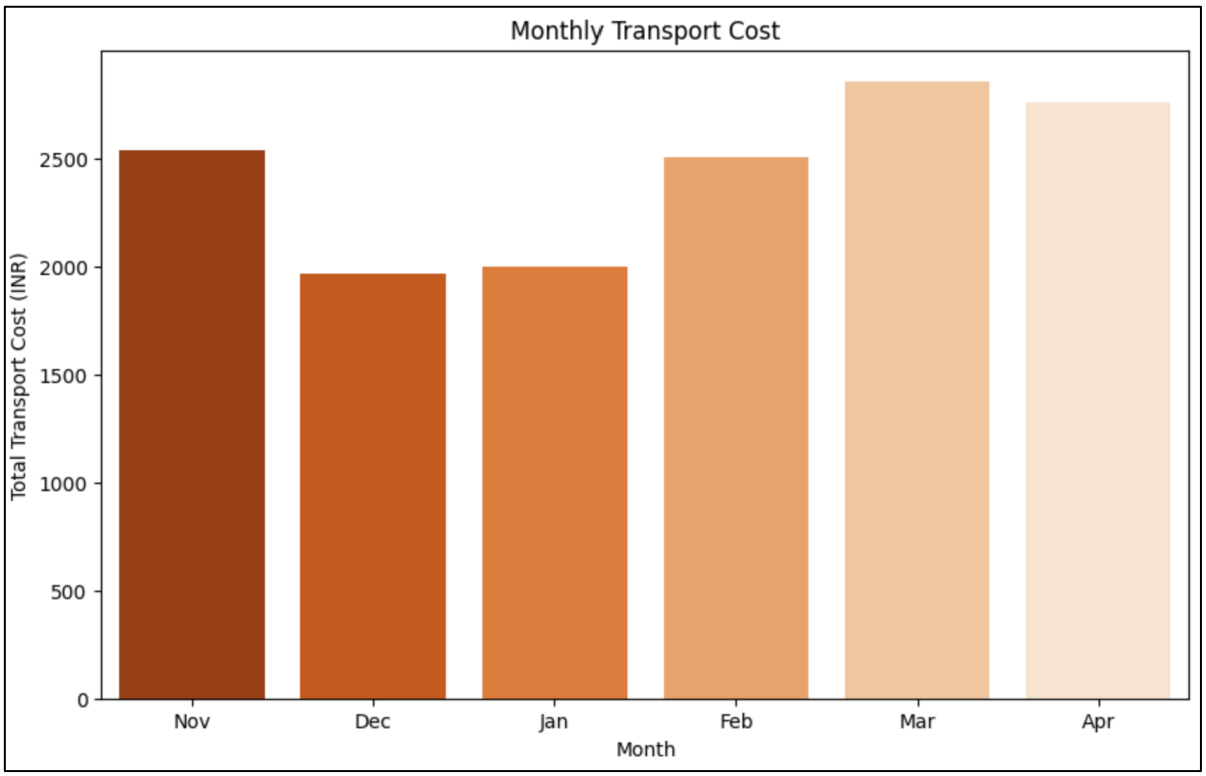

Fig 5 – Bar Chart: Monthly Transport Cost

- The line chart provides a detailed day-to-day view of transport costs throughout the period.
- Fluctuations and spikes show inconsistencies in shipment practices or rate changes.
- The highest recorded cost is ₹331, while the lowest drops to ₹0. This indicates possible data gaps or days of free shipping.
- The mean (about ₹103) and median (about ₹105) values are quite close, revealing a slightly right-skewed distribution but still balanced overall.
- March and April show more stable and compact patterns compared to the erratic behavior from November to January.

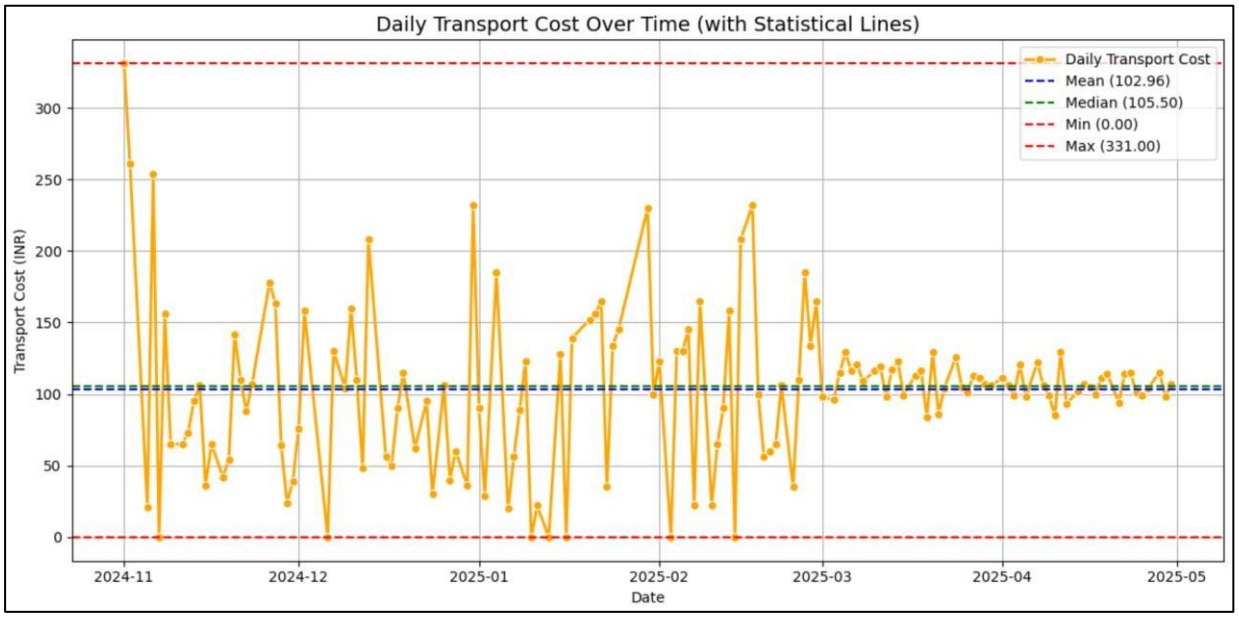

Fig 6 – Line Chart: Daily Transport Cost with Statistical Lines

3.4 Interdependency and Correlation Analysis

- Turmeric has the highest positive correlation with transport cost.
  This indicates that whenever Turmeric sales are high, there is a tendency for transportation cost to be high as well.
  This may be because Turmeric has greater weight or bulk orders than other spices, so logistics become costly.
- Both Dhaniya and Red Chilli also demonstrate a moderate positive correlation with transportation expenditures. This is to say that as their sales rise, transport costs do as well—but not as steeply as with Turmeric. This reinforces
  their position as necessary spices maintaining steady contributions to logistics activity.
- Cumin (–0.017), Ajwain (–0.16), and Fennel (–
  0.18) exhibit negative or weak correlations with transport.
  This suggests that fluctuations in their sales
  have minimal or no effect on the daily cost of transportation.
  These products may be sold

in small, infrequent units or may be packaged together with other spices in joint shipments.

- The matrix also indicates that the majority of spices are lowly correlated with each other, which reflects independent sales patterns.
  For example, increased sales of Dhaniya do not necessarily imply that Red Chilli or Fennel sales will go up as well. This piece of information is beneficial while creating promotions or demand forecasting for individual products.
- Since Turmeric, Red Chilli, and Dhaniya are the prime movers of transport expense, these must be given precedence for logistical optimization — e.g., route planning, load balancing, and cold chain management (particularly in case of perishable goods like Red Chilli).

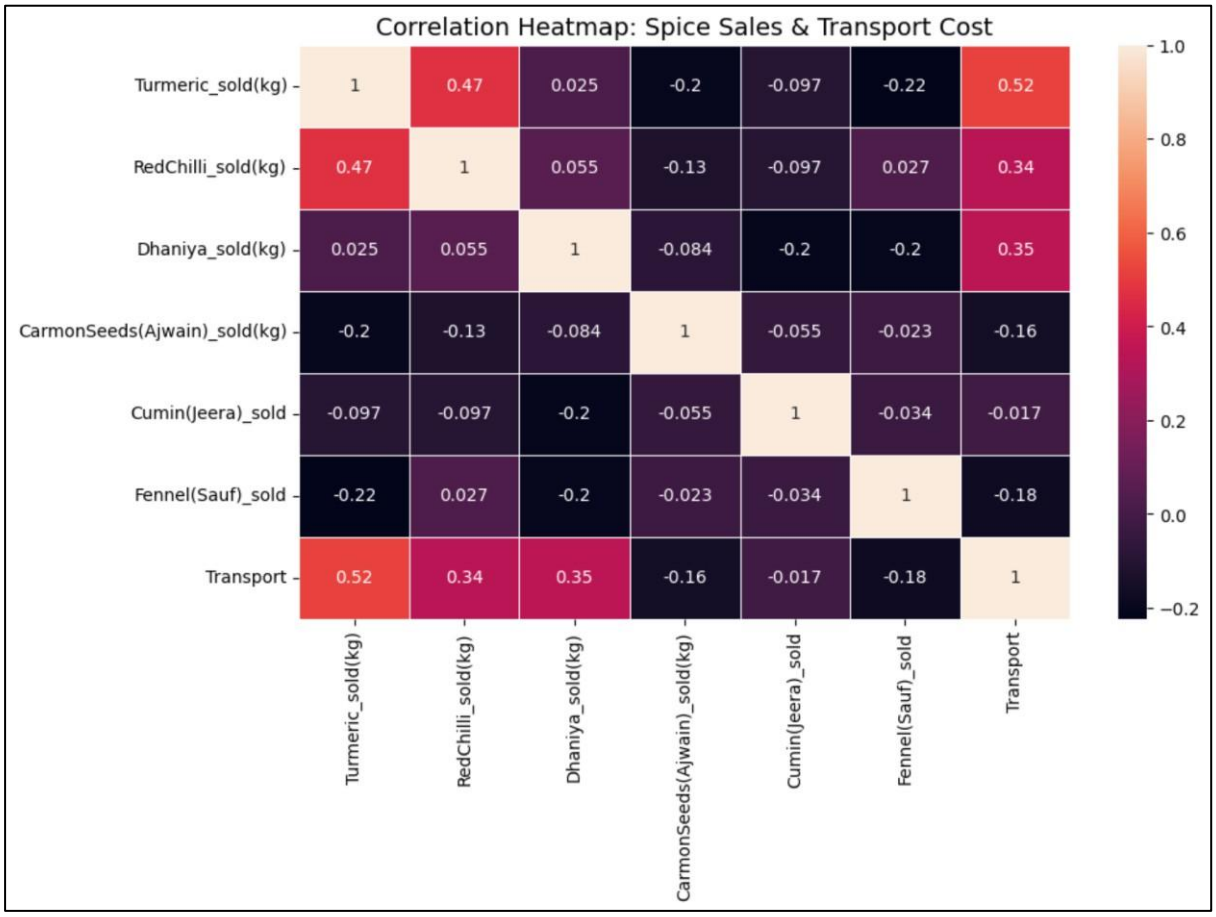

Fig 7 – Heatmap: Correlation Between Spice Sales & Transport Cost

4 Interpretation of Results and Recommendations

The main conclusions drawn from the data analyses and visualizations carried out during the study are fully interpreted in this section. Relevant charts and data-supported insights are used to support each identified issue, and then practical suggestions are made to address these problems. The recommendations seek to enhance perishable product handling, expedite logistics processes, and maximize inventory management.

4.1 Problem 1 – Overdependence on few Spices

4.1.1 Interpretation

- The Revenue Pie Chart reveals that Turmeric and Red Chilli account for 75.7% of the total revenue.
- ABC Analysis classifies Turmeric and Red Chilli as Category A items with remaining spices in B and C categories.
- This indicates that the business is heavily dependent on turmeric and red chilli for its earnings, which can create risk exposure in the long run.
- The business loses out on possible revenue streams from other spices that might have unrealized potential when it depends primarily on a small number of products. Ineffective use of capital, inventory space, and storage can also arise from this lack of variation in product performance.

4.1.2 Recommendations

- Diversify Revenue Streams -
  Increase the growth of underperforming spices like ajwain, fennel (saunf), and cumin (jeera), which have low revenue shares at the moment but could be raised with targeted marketing, shelf placement, or combo packaging. This spreads the risk over a larger range of products.
- Introduce Combo Offers and Bundling Strategies –
  Combine popular spices like turmeric with less popular products (such as combo packs of turmeric, cumin, and ajwain). Bundling can

raise average order value, decrease stagnant inventory, and generate interest in underperforming products.

- Track Emerging Consumer Demand –
  To see if demand is moving toward other products, use sales data and trend analysis. Sales of Dhaniya, for instance, increased in subsequent months; these indications can be used to determine which spices are likely to be future top sellers.

4.2 Problem 2 – Fluctuating Transportation Costs

4.2.1 Interpretation

- Daily Transport Line chart illustrates that the transport cost ranges from ₹0 to ₹331with noticeable daily fluctuations.
- Monthly transport cost bar chart shows significant increases in February and March which may be a sign of unforeseen dispatches or operational inefficiencies.
- While there's a positive trend, the scattered spread of data points around it reflects that transport costs do not increase in a linear or predictable manner with sales volume.
- All the analysis indicates that the transport costs are inconsistent and is not well-regulated. Costs fluctuate without any apparent pattern which suggests the business could either be using small or frequent deliveries, not consolidating loads, or using ad hoc transport.

4.2.2 Recommendations

- Implement Long-Term Vendor Contracts –
  Collaborating with a regular logistics provider through contracted monthly or per trip rates can help smoothen cost fluctuations and provide more control over transportation expenses.
- Route Optimization -
  Usage of Google Maps, Routific and other digital tools to batch deliveries going to the nearby areas. This reduces the number of trips, optimizes truckload efficiency and lowers the per-unit transport cost.

- Introduce Transport KPIs (Key Performance Indicators) and Monitoring –
  Establish benchmarks and monitor them on a monthly basis, such as the cost per kilogram sold, average delivery cost, or transport cost per order. This makes it possible to assess logistics performance over time using data.

4.3 Problem 3 – Spice Spoilage during Summer Months

4.3.1 Interpretation

- Monthly sales line chart shows the sharp decline of Red Chilli during the month of March and April. This deviation stands out and indicates the issue is specific to Red Chilli, rather than a general drop in demand across the business.
- According to the correlation heatmap, red chilli exhibits a moderately positive correlation (0.34) with the cost of transportation, indicating that it is transported frequently, perhaps in smaller quantities or with greater urgency.
- Given its generally high market demand, this decline is unlikely to be the result of a decline in consumer interest; rather, it might be the result of inventory losses brought on by heat-induced spoiling, breakage, or quality degradation.

4.3.2 Recommendations

- Seasonal Inventory Adjustment –
  During the summer, reduce the amount of red chili that is ordered or kept on hand. To better match inventory with what can actually be kept and sold during these dangerous times, use demand forecasting and spoilage history.
- Invest in Temperature-Controlled Storage –
  Use insulated bags or cold storage solutions for transportation, especially during the summer, to lower the possibility of spoiling. Even though there may be up-front expenses, this can drastically reduce waste and maintain product quality.

- Dynamic Pricing for Near Expiry Stock –
  As red chilli approaches its expiration or freshness limit, apply price reductions. Customers feel a sense of urgency as a result, which speeds up stock turnover and helps turn possible losses into profits.

5 Conclusion

Mahak Spices has a lot of potential, but operational inefficiencies are limiting its profitability and scalability, according to a six-month analysis of the business. Through thorough data exploration and visualization, key issues have been identified, including an excessive reliance on red chilli and turmeric, unstable transportation costs, and the summertime spoilage of perishable spices like red chilli.

The report reveals important information about product performance, logistical expenses, and sales trends by utilizing tools like Excel and Python. These results underline the necessity of better transport planning, more balanced inventory strategies, and enhanced management of sensitive stock during crucial times.

The operational efficiency of the company can be greatly increased by putting the suggested strategies into practice, such as adopting cold storage solutions, diversifying the product portfolio, and implementing route optimization. In the end, these data-driven will help Mahak Spices reduce losses, maximize profits, and lay a foundation for long-term, sustainable growth.